INCH-POUND

MIL-F-44251A
20 February 1990
SUPERSEDING
MIL-F-44251
20 November 1986

MILITARY SPECIFICATION

FRAME SECTIONS, TENT, EXTENDABLE, MODULAR, PERSONNEL (TEMPER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the frame sections of an extendable frame type personnel modular tent with covers.

1.2 Classification. The frame sections shall be of the following types and classes as specified (see 6.2):

Type I	-	Frame end section
Type II	-	Frame extendable section
Type III	-	Frame extendable door section
Class 1	-	Frame sections supplied with folding arch assemblies which allow a wrapped length of 113 inches.
Class 2	-	Frame sections supplied with sectionalized folding arch assemblies which allow a wrapped length of 98 inches

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8340

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SPECIFICATIONS

FEDERAL

DDD-L-20 - Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-P-53022 - Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-3330 - Frame Sections for Tent, Extendable, Modular, Personnel (TEMPER)
5-4-3331 - Binding Strap Assembly

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5-4-3332 - Side Arch Assembly
5-4-3333 - Roof Arch Assembly
5-4-3334 - Roof Arch Assembly, Ridge Gusset
5-4-3335 - Header Assembly
5-4-3336 - Purlin Assembly
5-4-3337 - Door Sill Assembly
5-4-3338 - Arch Assembly
5-4-3339 - Foot
5-4-3340 - Ridge Extender Assembly
5-4-3341 - Eave Extender Assembly
5-4-3342 - Hitch Pin Clip and Holder
5-4-3343 - Vestibule Frame Assembly
5-4-3344 - Upright
5-4-3345 - Header
5-4-3346 - Pin Assembly
5-4-3347 - Frame Sections Cover Assembly
5-4-3358 - Erection Instructions
5-4-4006 - Arch Assembly, Sectionalized
5-4-4007 - Roof Arch Assembly, Ridge Gusset, Sectionalized
5-4-4008 - Roof Arch Assembly, Sectionalized
5-4-4009 - Connector
5-4-4154 - Pin, Quick Release with Lanyard Assembly
5-4-4155 - Pin, Quick Release
5-4-4156 - Tab
5-4-4157 - Lanyard Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified herein and on the applicable drawings. Materials not definitely specified shall be of the quality normally used by the contractor provided that the completed item complies with all provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Primer. Primer shall conform to MIL-P-53022.

3.2.2 Coating. Coating shall be camouflage green 383 conforming to MIL-C-46168.

3.2.3 Thread, polyester. Dyed thread shall show colorfastness to weathering equal to or better than the standard shade sample. When no standard shade sample is available, the dyed thread shall show "good" colorfastness to weathering. Thread sizes shall be as follows:

For all stitching except bartacking and stitching indicated by "Z" on drawings	- Size FF or F for needle and bobbin
For bartacking	- Size E
For stitching indicated by "Z" on drawings	- Size F for needle, size E for looper

3.2.4 Buckle, double bar with lip. The double bar buckle with lip shall conform to the 1-inch, No. 789 steel Waterbury buckle or equal (see 6.5 and Drawing 5-4-3331).

3.3 Alternate components. Components shall be as specified on the applicable drawings and as specified herein. For components for which it is stated "or equal," if the contractor proposes to use an item considered to be equal to the component specified, prior to its use the contractor shall furnish a sample of the component with supporting data to the contracting officer for subsequent evaluation by the responsible military agency. The supporting data required shall prove the functional equivalence and design compatibility of the item proposed to be used. Prior to manufacture of the first article, or if none is

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required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of first article manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.4 Construction. Construction of the frame sections shall conform to the drawings listed in section 2 of this document and shall be as specified herein. The frame sections shall be capable of being erected and the stud brace shall be capable of being turned within the pad brace by hand without having to use a tool for force. The diagonal brace and shackle brace shall be capable of being rotated by hand. Heating of the hinge pin is allowed to facilitate upsetting the ends of the hinge pin securely into the countersink of the eave gusset plates so as to prevent the hinge pin from becoming loose and disengaged.

3.4.1 Welding. All welding shall be sound, smooth, uniform, and as specified.

3.4.2 Riveting. All rivets shall be securely set.

3.4.3 Castings. Castings shall be clean and free from pits, blowholes, and scale.

3.4.4 Stitching, machine.

3.4.4.1 Types of stitching. All stitch types, except bartacking, shall conform to FED-STD-751 as follows:

For all stitching except stitching indicated by "Z" on drawings	- Type 301, 5 to 7 stitches per inch
For stitching indicated by "Z" on drawings	- Type 401, 5 to 7 stitches per inch. Chain portion of stitching shall appear outside of covers
For bartacking	- Bartack, 5/8 plus or minus 1/16 inch long, 1/8 inch minimum bight, and shall contain 28 stitches

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3.4.4.2 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in puckering of material sewn. The lock shall be embedded in the material sewn.

3.4.4.2.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch behind the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch before the defective area (1/2 inch on box, box and cross and W-W stitching) and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.4.3 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of all seams or stitching produced with a 401 stitch type, when not turned under in a hem or held down by other stitching, shall have a 1/2 to 3/4 inch chain extending beyond each end. The looper (under thread) shall always be on the inside of the cover. All repairs shall be made in accordance with 3.4.4.2.1a or 3.4.4.2.1b. Repairs may be made using a 301 stitch type.

3.4.4.4 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping, or backstitches are used to secure the ends of stitching.

3.4.4.5 Lubrication of thread. The addition of any lubricant to the polyester thread prior to or during the sewing operation is prohibited.

3.5 Finish. Interior surfaces of rectangular tubes, where slotted or diagonally cut at end, and shank of through rivets need not be primed or painted. Flexible nylon coated cables and hitch pin chip assemblies need not be painted. Nonstructural rivets installed after treatment and painting need not be primed or painted (Ref: Drawing 5-4-3336, items 17 and 18, Drawing 5-4-3337, items 22 and 23, Drawing 5-4-4006, items 9 and 10, and Drawing 5-4-3338, items 9 and 10).

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3.5.1 Plating. All steel components on frame other than corrosion resistant steel shall be zinc-coated conforming to type II, SC3 of ASTM B 633 prior to treatment and coating unless otherwise specified on the drawings.

3.5.2 Cleaning and treatment. Prior to assembly, all completely fabricated aluminum components shall be thoroughly clean and the exterior surfaces of all aluminum shall be prepared for painting in accordance with class 1A treatment of MIL-C-5541. When a class 2 frame section is specified, the connectors for each sectionalized arch shall be treated.

3.5.3 Priming. After cleaning and treatment, as applicable, and prior to assembly, exterior surfaces of metal components including rings, screws, plus both sides of steel eave and ridge gusset-plates and header end plates, shall be completely coated with primer specified in 3.2.1. The shank of the quick release pin and connector (Drawing 5-4-4009) (class 2 only) when used shall be free of primer.

3.5.4 Coating. After the primer has been allowed to dry, the exterior surfaces of the assembled frame section components shall be completely coated with coating specified in 3.2.2. The shank of the quick release pin and connector (Drawing 5-4-4009) (class 2 only) shall be free of coating. The finish coating shall be continuous and uniform, e.g., free of sag, run, pinhole, blister, wrinkle, area of thin or no film, scratch or scuff, foreign matter imbedded in finish, and wet or tacky paint. The contractor shall touch up areas of finish damaged during flexing of movable joints.

3.6 Marking.

3.6.1 Indentification. The lower segment of each side arch assembly having the binding strap shall be permanently and distinctly stamped or marked for identification. The letters "U.S." shall be in characters a minimum of 1 inch in height. The name of the manufacturer or brand name or mark of such known character as to make the source of manufacture determinable and the number and year of contract shall be in characters a minimum of 1/2 inch in height. Identification marking shall conform to MIL-STD-130.

3.6.2 Marking for assembly. A white line measuring 3 inches by 1 inch shall be painted on top of each side of outside tube at connection on each class 2 roof arch segment to mark proper assembly of the arch. The coating shall conform to MIL-C-46168.

3.7 Erection instruction label. Labels of erection instructions shall conform to type VI, class 7 of DDD-L-20 and shall be stitched to the frame cover in the location shown on the drawing. The label shall be a proportionate copy of Drawing 5-4-3358 as reproduced from a glossy print of the drawing furnished by the contracting officer. The size label shall be, borderline to borderline, 15 inches $\pm$ 1/4 inches in the longer dimension, with a minimum 1/2 inch margin outside of the borderline on all sides.

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3.8 Workmanship. All components of the finished frame section shall not be bent out of shape and shall be free from split, puncture, fracture, twist, malformation, sharp edge, and burr. Roll of a rivet shall be free from more than one split. Welds shall also be free from burn through area, pits, slag, inclusion, porosity, crack, flash, and excess flux deposit. Rivets shall be free from looseness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3, 4.4.4, and 4.4.5.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies listed in table I shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

TABLE I. In-process inspection

Operation	Requirement paragraph
Welds are sound, smooth, uniform, and free from burn through area, pits, slag inclusion, porosity, crack, fracture, flash, and excess flux deposit	3.4.1 and 3.8
Castings are clean and free from pits, blowholes, and scale	3.4.3
There is no lubrication of polyester thread prior to or during sewing operations	3.4.4.5
All completely fabricated metal components are clean and exterior surfaces of metal are treated prior to assembly	3.5.2
Exterior surfaces of metal components including chains, rings, screws, plus both sides of steel eave and ridge gusset plates and header end plates (except shaft of quick release pin and connector) are primed before assembly	3.5.3
Primer is dry before applying coating	3.5.4

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of frame sections of one type and class only. The sample unit shall be one section with cover. The inspection level shall be I, and the acceptable quality level (AQL), shall be expressed in terms of defects per hundred units, 15 for major defects and 40 for total (major and minor combined) defects.

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color of finish is not as specified		201
	Shaft of quick release pin or connector has been coated with primer or coating		202
	Not continuous and not uniform, e.g., sag, pinhole, blister, wrinkle, foreign matter imbedded in finish, or wet or tacky finish		203
	Any run exceeding 3 inches in length	101	
	Area of no film:		
	-bare metal showing in area more than 3 square inches	102	
	-bare metal showing in area 3 square inches or less		204
	Sectionalized arch connector (for class 2) not treated as specified		205
	Note: Areas of no film on interior surfaces of rectangular tubes, where slotted or diagonally cut at end, and shank of through rivets are allowed. Areas of no film of 1/4 square inch or less are allowed on inside of boss, on purlin end fittings, and in area of brace pad adjacent to brace stud. Area of no enamel 1/16 inch wide is allowed at junction of riveted assemblies to tube.		
	Area of thin film:		
	-primer shows through in area 9 square inches or more	103	
	-primer shows through in area of less than 9 square inches		206
	Scratch or scuff exposing bare metal:		
	-12 inches or more in longest dimension	104	
	-less than 12 inches in longest dimension		207
	Note: Defects in finish attributable to flexing of movable joints during examination will not be scored as a defect. However, the contractor shall touch up areas of finish damaged during flexing of movable joints. Minor shade variations resulting from touch up operations shall not be considered a defect.		

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Fabric cover	Cut, tear, or hole (including exposed drill hole):		
	-greater than 1/8 inch in length or diameter	105	
	-1/8 inch or less in length or diameter:		
	-more than 10 holes but not more than 20 holes		208
	-more than 20 holes	106	
	Broken or missing yarn, multiple floats clearly visible at normal inspection distance (approximately 3 feet)		209
Tape and webbing	Edge frayed or scalloped		210
Hardware	Broken	107	
	Malformed:		
	-fails to perform intended function	108	
	-but will perform intended function		211
	Corroded area		212
	Burr or sharp edge which may cause injury in handling or damage to fabric	109	
Seams and stitching	Open seam on stitch type 401:		
	-one row of stitching open for 1 inch or less on cover		213
	-one row of stitching open for more than 1 inch on cover	110	
	-both rows of stitching open (not necessarily at same location) on cover	111	
	Open seam on stitch type 301		
	-more than 1/4 inch but not more than 1 inch		214
	-more than 1 inch	112	

Note: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or runoff stitches occur. On double stitched seams, a seam shall be classified as open when either one or both sides of seam are open.

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and Stitching (cont'd)	Raw edge caught in stitching:		
	-more than 1 inch but not more than 3 inches		215
	-more than 3 inches	113	
	Thread break:		
	-overstitched less than 1 inch on stitch type 401		216
	-overstitched less than 1 inch on stitch type 301		217
Note:	Thread breaks not overstitched shall be classified as open seams.		
	Skipped stitches:		
	-overstitched less than 1 inch on stitch type 401		218
	-overstitched less than 1 inch on stitch type 301 (when two or more skipped stitches occur consecutively)		219
Note:	Skipped stitches not overstitched shall be classified as open seams.		
	Needle chews resulting in cut, tear, or hole (see cut, tear, or hole)		
	One or more required rows of stitching omitted (except on boxstitching)	114	
Boxstitching	Incomplete:		
	-one row of stitching omitted		220
	-two or more rows of stitching omitted	115	
	Any boxstitching less than specified length by more than 1/8 inch		221
Stitching ends	Overstitched less than 1 inch on stitch type 301 (except where ends are held down by other stitching, turned under in a hem, or where stitching is performed automatically)		222
	Secured with less than three tying, overlapping, or backstitches when automatic stitching is performed		223

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams	Seam pleated		224
Seam type	Wrong seam type	116	
Stitch type	Wrong stitch type	117	
Stitch tension	Loose, resulting in an exposed bobbin or top thread, for more than 6 inches		225
	Tight, as evidenced by puckering on fabric, for more than 6 inches		226
Stitches per inch	Less than minimum specified:		
	-one stitch		227
	-two or more stitches	118	
	More than maximum specified:		
	-one stitch		228
	-two or more stitches	119	
Seams and stitching			
Stitching gage	Not as specified	120	
Stitching margin	Larger than specified by more than 3/16 inch:		
	-for more than 2 inches but not more than 4 inches in length		229
	-for more than 4 inches in length	121	
Stitching indicated by "Z" on drawings	Chain portion of stitching shows on outside of cover		230
Construction and workmanship	Any part or component not fabricated of applicable referenced materials	122	
	Any component part missing	123	
	Any component not fabricated as specified, e.g., stud not capable of being turned within the pad by hand without having to use a tool for force		231
	End of hinge pin not upset securely into the countersink of the eave gusset plate assembly, e.g., hinge pin is loose or disengaged	124	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Side arch becomes disassembled from roof arch due to disengagement of hinge pin	125	
	Any component split, punctured, fractured, twisted, malformed, or bent out of shape (except end plate of header assembly)	126	
	End plate of header assembly is bent out of shape to a degree that it will not function	127	
	End plate of header assembly is bent out of shape but will function		232
	Any sharp edge or burr	128	
Welding	Missing	129	
	Not sound; not smooth; not uniform; or has burn through area, pits, slag inclusion, porosity, crack, fracture, flash, or excess flux deposit		233
Riveting	Loose rivet		234
	Set of rivet is such that the diagonal brace or shackle brace is not capable of being rotated by hand operation	130	
	Roll of a rivet contains more than one split		235
Casting	Not clean or has pits, blowholes, or scale		236
Markings and labels	Omitted, illegible, incorrect, of improper size or type, or not in location specified		237

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of frame sections of one

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type and class only. The sample unit shall be one section with cover. The inspection level shall be I, and the AQL expressed in terms of defects per hundred unit is 15.

4.4.5 End item erection examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of frame sections of one type and class only. The sample unit shall be one section. Sections selected for visual or dimensional examination shall not be used for this examination. The inspection level shall be S-1, and the AQL expressed in terms of defects per hundred units is 2.5.

TABLE III. End item erection defects

Examine	Defect
Assembly	Any component or assembly that cannot be connected or joined with companion assembly, i.e., assembling requires realignment of parts, drilling, enlargement of holes, or use of drift pin or other mechanical means
	Assembly hole missing or misplaced
	Special tool is required to effect assembly
	Buckled, bent out of shape, or otherwise damaged
	Any component missing (check all clevis bolts, self locking nuts, and quick release pins)
	Component does not fit with mating components
	Component is out of alignment
Note:	The contractor shall touch up areas of finish damaged during erection and disassembly of the frame sections

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each combined side arch and roof arch assembly of the frame section (class 1 or class 2) shall have the foot plates wrapped, side arch wrapped at the joint of the roof arch and shall be completely folded and secured with the binding strap provided. The ends of the folded assembly shall be wrapped or cushioned to prevent damage to finish. The shackle brace end of the diagonal braces and the strap braces of each purlin shall be wrapped. The wrapped diagonal braces shall be folded toward each other and secured to the respective frame section purlins, or door sill assembly with the retaining straps. The pins of the header shall be secured inside the header end. The eave and ridge extenders shall be secured together with tape and wrapped or cushioned to prevent damage to finish. Vestibule frames shall be packaged in a component configuration that results in complete vestibule frames being packaged as units.

5.1.1.1 End section. One complete end section (two arches, class 1 or 2, two headers, five purlins, four eave extenders, and two ridge extenders), prepared as specified in 5.1.1, shall be nested in the most compact manner and placed in a frame sections cover and the cover secured by straps provided.

5.1.1.2 Extendable section. One complete extendable section (one arch, class 1 or 2, one header, five purlins, two eave extenders and one ridge extender), prepared as specified in 5.1.1, shall be placed in a frame sections cover as specified in 5.1.1.1.

5.1.1.3 Extendable door section. One complete extendable door section (one arch, class 1 or 2, one header, three purlins, two door sill assemblies, two vestibule frame assemblies, two eave extenders, and one ridge extender), prepared as specified in 5.1.1, shall be placed in a frame sections cover as specified in 5.1.1.1.

5.1.2 Commercial preservation. Components of each frame section shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One complete end section, or extendable section, or extendable door section, preserved as specified in 5.1, shall be packed in a close fitting shipping container conforming to overseas type, style A, type 3 load of PPP-B-601. Each box shall be provided with skids fabricated in accordance with the requirements of the box specification. Cushioning shall be used, as necessary, to prevent movement or damage during shipment. Each box shall be closed and reinforced with strapping in accordance with the appendix of PPP-B-601. The approximate inside dimensions of the shipping container shall be 114 inches in length, 20 inches in width, and 7-1/2 inches in depth for class 1 frame and 99 inches in length, 20 inches in width, and 7-1/2 inches in depth for class 2 frame.

5.2.1.1 Level A packing (when procured separately). Twenty arches, or 50 headers, or 64 purlins, or 480 eave extenders, or 480 ridge extenders, or 20 vestibule frame assemblies, preserved as specified in 5.1, shall be packed together in a close-fitting cleated plywood box conforming to overseas type, style A or B, type 3 load, of PPP-B-601. Cushioning shall be used, as necessary, to prevent movement or damage during shipment and storage. Each shipping container shall be closed and strapped as specified in the appendix of PPP-B-601.

5.2.2 Level B packing. One complete end section, or extendable section, or extendable door section, preserved as specified in 5.1, shall be packed as specified in 5.2.1 or 5.2.1.1, except that the box shall conform to class 2, style A, No. 1 or 2 end, of PPP-B-640. The approximate inside dimensions of the shipping container (for section assemblies only as described in 5.1.1.1, 5.1.1.2, and 5.1.1.3) shall be 114 inches in length, 20 inches in width, and 7-1/2 inches in depth for one class 1 frame and 99 inches in length, 20 inches in width, and 7-1/2 inches in depth for class 2 frame.

5.2.2.1 Vestibule frame assembly. One vestibule frame assembly, preserved as specified in 5.1, shall be packed in a close fitting fiberboard container conforming to style FTC, grade V3c of PPP-B-636. The approximate inside dimensions of the shipping container shall be 79-1/2 inches in length, 13-1/2 inches in width, and 3-1/2 inches in depth. The fiberboard container shall be closed as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Frame sections, vestibule frame assembly and tent frame covers or individual components when procured separately, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The frame sections are intended for use in the TEMPER TENT as Structural, Extendable Frame Sections. One end section will provide support for a tent 8 feet in length. The frame can be lengthened in 8 foot increments by the addition of either extendable sections or extendable door sections.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Drawings. The drawings applicable to class 1 and class 2 frames are as follows:

Class 1		Class 2	
5-4-3330	5-4-3331	5-4-3330	5-4-3331
5-4-3332	5-4-3333	5-4-3332	5-4-3335
5-4-3334	5-4-3335	5-4-3336	5-4-3337
5-4-3336	5-4-3337	5-4-3339	5-4-3340
5-4-3338	5-4-3339	5-4-3341	5-4-3342
5-4-3340	5-4-3341	5-4-3343	5-4-3344
5-4-3342	5-4-3343	5-4-3345	5-4-3346
5-4-3344	5-4-3345	5-4-3347	5-4-3358
5-4-3346	5-4-3347	5-4-4006	5-4-4007
5-4-3358	5-4-4154	5-4-4008	5-4-4009
5-4-4155	5-4-4156	5-4-4154	5-4-4155
5-4-4157		5-4-4156	5-4-4157

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6.5 Commercial product. The Waterbury buckle No. 789 manufactured by the Waterbury Buckle Co., Waterbury, CT, has been found to meet the requirements specified in 3.2.4.

6.6 Subject term (key word) listing.

End sections
Extendable sections
Extendable door sections
Sections with arch assemblies
TEMPER

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-44251A	FRAME SECTIONS, TENT, EXTENDABLE, MODULAR, PERSONNEL (TEMPER)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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